

FGFR2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR2 gene fusions from the msk_impact_50k_2026 study, covering 136 fusion events. 108/136 fusions (79.4%) were in-frame. 115/136 fusions (84.6%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.000424687$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain, $p=0.000424687$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: FGFR2 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 134 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 732 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -25 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: FGFR2 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BICC1	28	0.20588235294117646	0.20043213737826426	None	0.9354064039408867	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.3602624297400743	1
TACC2	8	0.058823529411764705	0.20043213737826426	None	0.9314655172413793	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.29639274393471765	2
ANKRD11	1	0.007352941176470588	0.20043213737826426	None	1.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28901988099643894	3
NRG3	2	0.014705882352941176	0.20043213737826426	None	0.9698275862068966	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2857056242592613	4
NOL4	4	0.029411764705882353	0.20043213737826426	None	0.9392241379310344	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2854502634942656	5
KIAA1217	4	0.029411764705882353	0.20043213737826426	None	0.9387931034482758	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28535789896224595	6
TBC1D4	3	0.022058823529411766	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28239136752208366	7
CTNNA3	3	0.022058823529411766	0.20043213737826426	None	0.9390804597701149	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28226821481272407	8
SHTN1	3	0.022058823529411766	0.20043213737826426	None	0.9385057471264368	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28214506210336443	9
INA	2	0.014705882352941176	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	10
SORBS1	2	0.014705882352941176	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	11
TTC28	2	0.014705882352941176	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	12
WAC	2	0.014705882352941176	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	13
AHCYL1	2	0.014705882352941176	0.20043213737826426	None	0.9387931034482758	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2790553779538426	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BAIAP2L1	2	0.014705882352941176	0.20043213737826426	None	0.9387931034482758	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2790553779538426	15
NRBF2	2	0.014705882352941176	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2788706488898032	16
OPTN	2	0.014705882352941176	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2788706488898032	17
ABLIM1	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	18
ADAM9	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	19
AMOT	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	20
ATAD1	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	21
ATP6V1D	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	22
BICD1	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	23
BNIP2	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	24
CAT	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	25
CCDC6	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	26
CEACAM7	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	27
CEP112	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	28
DDX21	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	29
DIABLO	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	30

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
GAB2	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	31
HSPA12A	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	32
KLHL29	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	33
MARVELD3	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	34
MYO18B	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	35
NRAP	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	36
PDZRN4	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	37
PKP4	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	38
POC1B	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	39
PRDX3	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	40
RNF145	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	41
RPAP3	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	42
SH3GLB1	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	43
SHC2	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	44
SKI	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	45
SRRT	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	46

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TFEC	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	47
VCL	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	48
VPS26A	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	49
YTHDF3	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	50
ZMYM4	1	0.007352941176470588	0.20043213737826426	None	0.9396551724137931	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	51
BTBD16	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	52
CCDC186	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	53
CCDC73	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	54
KIF14	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	55
L3MBTL3	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	56
LAMC1	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	57
MBIP	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	58
MYH15	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	59
MYH9	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	60
MYLK	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	61
NID2	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	62

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PAH	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	63
PALD	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	64
PPP1R21	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	65
RABGAP1L	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	66
ROCK1	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	67
SEPTIN10	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	68
STAT4	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	69
UBP1	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	70
ZFAT	1	0.007352941176470588	0.20043213737826426	None	0.9379310344827586	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	71
FOXP1	2	0.014705882352941176	0.20043213737826426	None	0.9103448275862069	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27295931884054214	72
ADAM12	1	0.007352941176470588	0.20043213737826426	None	0.903448275862069	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2683302258240251	73
KCNH7	1	0.007352941176470588	0.20043213737826426	None	0.8862068965517241	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26463564454323696	74
ATE1	3	0.022058823529411766	0.20043213737826426	None	0.8080459770114943	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2541893970787339	75
PRKAG2	1	0.007352941176470588	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.09511621217728589	76
ST8SIA6	1	0.007352941176470588	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.09511621217728589	77
GRID1	1	0.007352941176470588	0.20043213737826426	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07473416671072464	78

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
152	0	115	1	18	0.0	0.1417910447761194	0.8483511974119787
469	0	115	2	17	0.0	0.019189765458422176	1.7169303332757584
676	0	115	3	16	0.0	0.0024714091878270984	2.607055343103334
732	0	115	5	14	0.0	3.4829019675778254e-05	4.458058749354329
767	63	52	15	4	0.3230769230769231	0.07677359623872276	1.1147881156383952
768	111	4	19	0	0.0	1.0	-0.0
798	112	3	19	0	0.0	1.0	-0.0
801	113	2	19	0	0.0	1.0	-0.0
802	114	1	19	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

99	16
9	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	134	115	3	16	0.09006270188105643

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABLIM1	1
ADAM12	1
ADAM9	1
AHCYL1	2
AMOT	1
ANKRD11	1
ATAD1	1

Partner_gene	Event_count
ATE1	3
ATP6V1D	1
BAIAP2L1	2
BICC1	28
BICD1	1
BNIP2	1
BTBD16	1
CAT	1
CCDC186	1
CCDC6	1
CCDC73	1
CEACAM7	1
CEP112	1
CTNNA3	3
DDX21	1
DIABLO	1
FOXP1	2
GAB2	1
GRID1	1
HSPA12A	1
INA	2
KCNH7	1
KIAA1217	4
KIF14	1
KLHL29	1
L3MBTL3	1
LAMC1	1
MARVELD3	1
MBIP	1
MYH15	1
MYH9	1
MYLK	1
MYO18B	1
NID2	1
NOL4	4

Partner_gene	Event_count
NRAP	1
NRBF2	2
NRG3	2
OPTN	2
PAH	1
PALLD	1
PDZRN4	1
PKP4	1
POC1B	1
PPP1R21	1
PRDX3	1
PRKAG2	1
RABGAP1L	1
RNF145	1
ROCK1	1
RPAP3	1
SEPTIN10	1
SH3GLB1	1
SHC2	1
SHTN1	3
SKI	1
SORBS1	2
SRRT	1
ST8SIA6	1
STAT4	1
TACC2	8
TBC1D4	3
TFEC	1
TTC28	2
UBP1	1
VCL	1
VPS26A	1
WAC	2
YTHDF3	1
ZFAT	1

Partner_gene	Event_count
ZMYM4	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshifts	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 1826-T01-I M6-1	P-0021 826-T0 1-IM6		ANKRD11-FGFR2	ANKRD11	unknown	five_prime	123243799	17	732	True	disrupted	0.9097472924187726	True			Protein tyrosine and serine/threonine kinase	FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10;16)(q26.13;q24.3)(chr10:g.123243799::chr16:g.89538175) Transcript Fusion {FGFR2:ANKRD11}	NA
EVT-P-001 3241-T01-I M5-2	P-0013 241-T0 1-IM5		ATP6V1D-FGFR2	ATP6V1D	in-frame	five_prime	123242048	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - ATP6V1D (NM_015994) rearrangement: t(10;14)(q26.13;q23.2)(chr10:g.123242048::chr14:g.67818116) Protein fusion: in frame (FGFR2-ATP6V1D)	NA
EVT-P-000 9512-T01-I M5-3	P-0009 512-T0 1-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123240449	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Note: The FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal rearrangement event is an inversion which results in the fusion of FGFR2 exons 1-17 with BICC1 exons 3-21. The fusion is predicted to be in frame and includes the kinase domain of FGFR2. Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-000 2675-T01-I M3-4	P-0002 675-T0 1-IM3		BICC1-FGFR2	BICC1	in-frame	five_prime	123241096	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-002 9067-T01-I M6-5	P-0029 067-T0 1-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123240197	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21) : c.2302-67:FGFR2_c.238-6757:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003177-T01-I M5-6	P-0003177-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123242188	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0025807-T01-I M6-7	P-0025807-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123242181	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0025250-T01-I M6-8	P-0025250-T01-IM6		BICC1-FGFR2	BICC1	in-frame	three_prime	123240491	18	768	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 16-21): c.2302-956:FGFR2_c.2182-297:BICC1inv Protein Fusion: in frame {BICC1:FGFR2}	NA
EVT-P-0032033-T01-I M6-9	P-0032033-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241685	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0005538-T01-I M5-10	P-0005538-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123242753	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:FGFR2_c.2182-276:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0020186-T01-I M6-11	P-0020186-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241148	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exon17 fused with BICC1 exon3): c.2302-1613:FGFR2_c.237+32254: BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 0025-T01-I M6-12	P-0020 025-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	fiv e_ pri me	123241 984	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 3-21) : c.2 301+1228:FGFR2_c.238-47 73:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-001 8115-T01-I M6-13	P-0018 115-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	fiv e_ pri me	123241 970	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused in-frame with BICC1 exons 3-21) : c.2301+1242:F GFR2_c.237+34947:BICC1d up Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-001 5942-T01-I M6-14	P-0015 942-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	fiv e_ pri me	123241 020	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exons 1-17 with BICC1 exons 9-21) : c.2 302-1485:FGFR2_c.1048-60 :BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-000 8146-T01-I M5-15	P-0008 146-T0 1-IM5		BICC1- FGFR2	BIC C1	in-fr ame	fiv e_ pri me	123240 154	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2302-619:FGFR2_ c.2182-744:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-003 7617-T01-I M6-16	P-0037 617-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	thr ee_ pri me	123241 302	18	768	True	lost	0.0	False		Prote in tyr osine and serin e/thr eonine ki nase		BICC1 (NM_001080512) - FGFR2 (NM_000141) rearrangement: c.238-34060: BICC1_c.2302-1767:FGFR2i nv Protein Fusion: in frame {BICC1:FGFR2}	NA
EVT-P-001 5432-T01-I M6-17	P-0015 432-T0 1-IM6		CAT-F GFR2	CAT	in-fr ame	fiv e_ pri me	123242 534	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26. 13;p13)(chr10:g.123242534:: chr11:g.34467980) Protein fusion: in frame (FGFR2-CAT)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 7305-T01-I M6-18	P-0027 305-T0 1-IM6		CCDC 6-FGF R2	CC DC6	in-fr ame	fiv_e_prime	123241 542	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+ 1670:FGFR2_c.582+6426:C CDC6del Protein Fusion: in frame {FGFR2:CCDC6}	NA
EVT-P-002 7741-T01-I M6-19	P-0027 741-T0 1-IM6		CEAC AM7-F GFR2	CEA CA M7	unk now n	fiv_e_prime	123241 990	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) rearrangement: t(10;19)(q26. 13;q13.2)(chr10:g.12324199 0::chr19:g.42175729) Transcript Fusion {FGFR2:CEACAM7}	NA
EVT-P-000 3724-T01-I M5-20	P-0003 724-T0 1-IM5		CEP11 2-FGF R2	CEP 112	in-fr ame	thre_e_prime	123241 923	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eonine ki nase		t(17;10)(q24.1;q26.13)(chr17 :g.64023428::chr10:g.12324 1923) Protein fusion: in frame (CCDC46-FGFR2)	NA
EVT-P-001 9723-T01-I M6-22	P-0019 723-T0 1-IM6		CTNN A3-FG FR2	CTN NA3	in-fr ame	fiv_e_prime	123243 018	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194: FGFR2_c.1885-25982:CTN NA3del Protein Fusion: in frame {FGFR2:CTNNA3}	NA
EVT-P-001 1516-T01-I M5-23	P-0011 516-T0 1-IM5		CTNN A3-FG FR2	CTN NA3	in-fr ame	thre_e_prime	123241 940	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eonine ki nase		CTNNA3 (NM_013266) -FGFR2 (NM_000141) Rearrangement : c.1048-847 9:CTNNA3_c.2301+1272:FG FR2dup Protein fusion: in frame (CTNNA3-FGFR2)	NA
EVT-P-002 0410-T01-I M6-24	P-0020 410-T0 1-IM6		DDX21 -FGFR 2	DD X21	in-fr ame	fiv_e_prime	123243 048	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - DDX21 (NM_004728) fusion (FGFR2 exons 1-17 fused to DDX21 exons 2-15): c.2301 +164:FGFR2_c.88-860:DDX 21inv Protein Fusion: in frame {FGFR2:DDX21}	NA
EVT-P-001 8869-T03-I M6-26	P-0018 869-T0 3-IM6		FGFR2 -ABLI M1	ABL IM1	in-fr ame	fiv_e_prime	123242 601	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - ABLIM1 (NM_002313) fusion: c.2301+611:FGFR2_ c.895-12244:ABLIM1del Protein Fusion: in frame {FGFR2:ABLIM1}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 6275-T01-I M6-27	P-0026 275-T01-IM6		FGFR2-ADAM12	ADAM12	unkn_ow_n	fiv_e_pri_me	123246899	15	676	False	disrupt_	0.7075812274368231	True			Protein tyrosin_e and s_erine/threonine kinase	FGFR2 (NM_000141) - ADAM12 (NM_003474) rearrangement: c.2026:FGFR2_c.1618+5260:ADAM12dup Protein Fusion: mid-exon {FGFR2:ADAM12}	NA
EVT-P-002 0164-T01-I M6-28	P-0020 164-T01-IM6		FGFR2-ADAM9	ADAM9	in-fr_ame	fiv_e_pri_me	123242738	17	767	True	retain_	1.0	False	Protei_n tyrosine and se_rine/threonine kinase			FGFR2 (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10(p11.22;q26.13)(chr8:g.38901161::chr10:g.123242738) Protein Fusion: in frame {FGFR2:ADAM9}	NA
EVT-P-001 8920-T01-I M6-29	P-0018 920-T01-IM6		FGFR2-AHCYL1	AHCYL1	in-fr_ame	fiv_e_pri_me	123242706	17	767	True	retain_	1.0	False	Protei_n tyrosine and se_rine/threonine kinase			FGFR2 (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AHCYL1 exons 2-17): t(1;10)(p13.3;q26.13)(chr1.g.110531185::chr10:g.123242706) Protein Fusion: in frame {FGFR2:AHCYL1}	NA
EVT-P-000 0644-T01-I M3-30	P-0000 644-T01-IM3		FGFR2-AHCYL1	AHCYL1	in-fr_ame	thre_e_pr_im_e	123239745	18	768	True	lost	0.0	False		Prote_in tyrosine and serin_e/threoni_ne ki_nase		NA Protein fusion: in frame (AHCYL1-FGFR2)	NA
EVT-P-001 2666-T01-I M5-31	P-0012 666-T01-IM5		FGFR2-AMOT	AMOT	in-fr_ame	fiv_e_pri_me	123241762	17	767	True	retain_	1.0	False	Protei_n tyrosine and se_rine/threonine kinase			FGFR2 (NM_000141) - AMOT (NM_001113490) Rearrangement : t(10;X)(q26.13.1;q23)(chr10:g.123241762::chrX:g.112058354) Protein fusion: in frame (FGFR2-AMOT)	NA
EVT-P-004 7504-T01-I M6-32	P-0047 504-T01-IM6		FGFR2-ATAD1	ATAD1	in-fr_ame	fiv_e_pri_me	123241680	17	767	True	retain_	1.0	False	Protei_n tyrosine and se_rine/threonine kinase			FGFR2 (NM_000141) - ATAD1 (NM_032810) fusion: c.2301+1532:FGFR2_c.831+3993:ATAD1del Protein Fusion: in frame {FGFR2:ATAD1}	NA
EVT-P-001 7424-T01-I M6-33	P-0017 424-T01-IM6		FGFR2-ATE1	ATE1	out-of-fr_ame	fiv_e_pri_me	123240443	18	768	True	retain_	1.0	False	Protei_n tyrosine and se_rine/threonine kinase			FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908:FGFR2_c.1379-520:ATE1dup Protein Fusion: out of frame {FGFR2:ATE1}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_st_atu_s	tar_get_r_ole	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 3512-T01-I M7-35	P-0063 512-T0 1-IM7		FGFR2 -ATE1	ATE 1	unk now n	fiv e_ pri me	123263 336	10	469	False	lost	0.0	False		Prote in tyro sine and serine/thr eonine ki nase		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGF R2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1}	NA
EVT-P-006 3625-T01-I M7-36	P-0063 625-T0 1-IM7		FGFR2 -ATE1	ATE 1	in-fr ame	fiv e_ pri me	123242 847	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_ c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1}	NA
EVT-P-005 2794-T01-I M6-38	P-0052 794-T0 1-IM6		FGFR2 -BAIAP 2L1	BAI AP2 L1	in-fr ame	fiv e_ pri me	123240 813	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::chr10:g.12 3240813) Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA
EVT-P-005 9296-T01-I M7-39	P-0059 296-T0 1-IM7		FGFR2 -BAIAP 2L1	BAI AP2 L1	in-fr ame	fiv e_ pri me	123242 027	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::chr10:g.12 3242027) Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA
EVT-P-000 4520-T01-I M5-41	P-0004 520-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 269	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2(NM_000141) - BICC1(NM_001080512) Fusion (FGFR2 exon 17 fused with exon 10 of BICC1) : c.2302-734:FGFR2_c.1180 -924:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-001 3174-T01-I M5-42	P-0013 174-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 123	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c. 2302-588:FGFR2_c.2182-28 4:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-004 9135-T07-I M7-43	P-0049 135-T0 7-IM7		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123239 791	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine ki nase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c .238-36290:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 8640-T01-I M7-44	P-0068 640-T0 1-IM7		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123240 444	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-909:FGFR2_c.238-18504:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-004 9135-T03-I M6-45	P-0049 135-T0 3-IM6		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123239 791	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c.238-36290:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-004 9133-T04-I M6-46	P-0049 133-T0 4-IM6		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123242 098	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2_c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-000 6885-T01-I M5-48	P-0006 885-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123240 587	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) rearrangement: c.2302-1052:FGFR2_c.238-5239:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-005 1773-T01-I M6-49	P-0051 773-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123242 828	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+384:FGFR2_c.237+13961:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-004 9133-T02-I M6-50	P-0049 133-T0 2-IM6		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123242 098	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2_c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-005 3822-T01-I M6-51	P-0053 822-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-frame	five_prime	123243 033	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+179:FGFR2_c.238-1932:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 1327-T01-I M6-52	P-0021 327-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 344	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-1613:FGFR2 _c.237+32254:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-006 0245-T01-I M7-53	P-0060 245-T0 1-IM7		FGFR2 -BICC1	BIC C1	unk now n	fiv e_ pri me	123239 377	18	820	False	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.117 9+1103:BICC1inv Protein Fusion: mid-exon {FGFR2:BICC1}	NA
EVT-P-006 8033-T01-I M7-54	P-0068 033-T0 1-IM7		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123239 673	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c .238-40422:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-003 9087-T01-I M6-55	P-0039 087-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123242 059	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2301+1153:FGFR 2_c.2181+133:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-005 4191-T01-I M6-56	P-0054 191-T0 1-IM6		FGFR2 -BICD1	BIC D1	in-fr ame	fiv e_ pri me	123242 849	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BICD1 (NM_001714) fusion: t(10;12)(q26.13;p11.21)(chr1 0:g.123242849::chr12:g.324 65127) Protein Fusion: in frame {FGFR2:BICD1}	NA
EVT-P-006 1764-T01-I M7-57	P-0061 764-T0 1-IM7		FGFR2 -BNIP2	BNI P2	in-fr ame	fiv e_ pri me	123243 170	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BNIP2 (NM_004330) fusion: t(10;15)(q26.13;q22.2)(chr10 :g.123243170::chr15:g.5997 5666) Protein Fusion: in frame {FGFR2:BNIP2}	NA
EVT-P-002 0135-T01-I M6-58	P-0020 135-T0 1-IM6		FGFR2 -BTBD 16	BTB D16	out-of-fr ame	fiv e_ pri me	123239 702	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - BTBD16 (NM_144587) rearrangement: c.2302-167: FGFR2_c.386-107: BTBD16inv Protein Fusion: out of frame {FGFR2:BTBD16}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_st_atu_s	tar_get_r_ole	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 3652-T01-I M6-59	P-0033 652-T0 1-IM6		FGFR2 -CCDC 186	CC DC1 86	in-fr ame	fiv e_ pri me	123240 830	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CCDC186(NM_018017) Rearrangement : c.2302-129 5:FGFR2_c.1222-3356:CCD C186del Protein Fusion: in frame {FGFR2:CCDC186}	NA
EVT-P-006 0582-T01-I M7-60	P-0060 582-T0 1-IM7		FGFR2 -CCDC 73	CC DC7 3	in-fr ame	fiv e_ pri me	123241 248	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CCDC73 (NM_001008391) fusion: t(10;11)(q26.13;p13)(chr10:g.123241248::chr11:g. 32747346) Protein Fusion: in frame {FGFR2:CCDC73}	NA
EVT-P-003 7574-T01-I M6-61	P-0037 574-T0 1-IM6		FGFR2 -CTNN A3	CTN NA3	in-fr ame	fiv e_ pri me	123240 911	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - CTNNA3 (NM_013266) fusion: c.2302-1376:FGFR2_ c.1884+30653:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3}	NA
EVT-P-004 4768-T01-I M6-63	P-0044 768-T0 1-IM6		FGFR2 -DIABL O	DIA BLO	in-fr ame	fiv e_ pri me	123243 152	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - DIABLO (NM_019887) fusion: t(10;12)(q26.13;q24.3 1)(chr10:g.123243152::chr12 :g.122708063) Protein Fusion: in frame {FGFR2:DIABLO}	NA
EVT-P-005 8539-T01-I M6-101	P-0058 539-T0 1-IM6		FGFR2 -FOXP 1	FOX P1	unk now n	fiv e_ pri me	123239 434	18	801	False	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(c hr3:g.71113500::chr10:g.123 239434) Protein Fusion: mid-exon {FGFR2:FOXP1}	NA
EVT-P-006 1783-T01-I M7-102	P-0061 783-T0 1-IM7		FGFR2 -FOXP 1	FOX P1	in-fr ame	fiv e_ pri me	123241 483	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(c hr3:g.71192573::chr10:g.123 241483) Protein Fusion: in frame {FGFR2:FOXP1}	NA
EVT-P-005 5215-T01-I M6-103	P-0055 215-T0 1-IM6		FGFR2 -GRID 1	GRI D1	out- of-fr ame	fiv e_ pri me	123323 934	4	152	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase		FGFR2 (NM_000141) - GRID1 (NM_017551) rearrangement: c.454+82:FG FR2_c.726+103661:GRID1d el Protein Fusion: out of frame {FGFR2:GRID1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 5437-T01-I M6-104	P-0015 437-T0 1-IM6		FGFR2 -HSPA 12A	HSP A12 A	in-frame	five_prime	123242 591	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - HSPA12A (NM_025015) rearrangement: c.2301+621:FGFR2_c.663+3346:HSPA12A Protein fusion: in frame (FGFR2-HSPA12A)	NA
EVT-P-000 9701-T01-I M5-105	P-0009 701-T0 1-IM5		FGFR2 -KCNH 7	KC NH7	unknown	five_prime	123239 444	18	798	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KCNH7 (NM_033272) Rearrangement : t(2;10)(q24.3; q26.13) (chr2:g.163263969::chr10:g.123239444) Protein fusion: mid-exon (FGFR2-KCNH7)	NA
EVT-P-000 8658-T01-I M5-107	P-0008 658-T0 1-IM5		FGFR2 -KIAA1 217	KIA A12 17	in-frame	five_prime	123241 193	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2302-1658:FGFR2_c.355-22274:KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-003 4784-T01-I M6-108	P-0034 784-T0 1-IM6		FGFR2 -KIAA1 217	KIA A12 17	in-frame	five_prime	123242 254	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1217 (NM_019590) fusion: c.2301+958:FGFR2_c.355-49659:KIAA1217inv Protein Fusion: in frame {FGFR2:KIAA1217}	NA
EVT-P-001 3405-T01-I M5-109	P-0013 405-T0 1-IM5		FGFR2 -KIAA1 217	KIA A12 17	in-frame	five_prime	123241 766	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2301+1446:FGFR2_c.354+73179:KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-000 3660-T01-I M5-110	P-0003 660-T0 1-IM5		FGFR2 -KIF14	KIF 14	in-frame	five_prime	123240 038	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			t(1;10)(q32.1;q26.13)(chr1:g.200555931::chr10:g.123240038) Protein fusion: in frame (FGFR2-KIF14)	NA
EVT-P-001 3241-T01-I M5-111	P-0013 241-T0 1-IM5		FGFR2 -KLHL 29	KLH L29	unknown	three_prime	123241 923	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		KLHL29 (NM_052920) - FGFR2 (NM_000141) rearrangement: t(2;10)(p24.1;q26.13)(chr2:g.23647979::chr10:g.123241923) Transcript fusion (KLHL29-FGFR2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 9439-T02-I M6-112	P-0019 439-T0 2-IM6		FGFR2 -L3MB TL3	L3MBTL3	in-frame	five_prime	123239 680	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - L3MBTL3 (NM_032438) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) Protein Fusion: in frame {FGFR2:L3MBTL3}	NA
EVT-P-001 1394-T01-I M5-113	P-0011 394-T0 1-IM5		FGFR2 -LAMC 1	LAMC1	in-frame	five_prime	123240 042	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - LAMC1 (NM_002293) rearrangement: t(1;10)(q25.3;q26.13)(chr1:g.183109471::chr10:g.123240042) Protein fusion: in frame {FGFR2-LAMC1}	NA
EVT-P-006 4855-T01-I M7-115	P-0064 855-T0 1-IM7		FGFR2 -MBIP	MBIP	in-frame	five_prime	123240 070	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - MBIP (NM_016586) fusion: t(10;14)(q26.13;q13.3)(chr10:g.123240070::chr14:g.36787130) Protein Fusion: in frame {FGFR2:MBIP}	NA
EVT-P-003 0844-T01-I M6-116	P-0030 844-T0 1-IM6		FGFR2 -MYH1 5	MYH15	in-frame	five_prime	123240 665	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.1;q26.13)(chr3:g.108134575::chr10:g.123240665) Protein Fusion: in frame {FGFR2:MYH15}	NA
EVT-P-002 7623-T01-I M6-117	P-0027 623-T0 1-IM6		FGFR2 -MYLK	MYLK	unknown	five_prime	123240 959	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612::chr10:g.123240959) Transcript Fusion {FGFR2:MYLK}	NA
EVT-P-003 8235-T01-I M6-118	P-0038 235-T0 1-IM6		FGFR2 -NOL4	NOL4	in-frame	five_prime	123240 418	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418::chr18:g.31580817) Protein Fusion: in frame {FGFR2:NOL4}	NA
EVT-P-001 2423-T02-I M6-119	P-0012 423-T0 2-IM6		FGFR2 -NOL4	NOL4	in-frame	five_prime	123242 055	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123242055::chr18:g.31563418) Protein Fusion: in frame {FGFR2:NOL4}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 6328-T02-I M6-120	P-0006 328-T0 2-IM6		FGFR2 -NRBF 2	NR BF2	in-fr ame	fiv e_ pri me	123241 246	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157- 363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2}	NA
EVT-P-005 8339-T01-I M6-121	P-0058 339-T0 1-IM6		FGFR2 -PAH	PAH	in-fr ame	fiv e_ pri me	123240 026	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)q26.13;q23.2)(chr10:g.123 240026::chr12:g.103294473) Protein Fusion: in frame {FGFR2:PAH}	NA
EVT-P-001 5500-T01-I M6-122	P-0015 500-T0 1-IM6		FGFR2 -PALL D	PAL LD	in-fr ame	fiv e_ pri me	123239 869	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - PALLD (NM_001166108) Rearrangement : t(4;10)(q33, q26.13)(chr4:g.169605121::c hr10:g.123239869) Protein fusion: in frame (FGFR2-PALLD)	NA
EVT-P-002 1448-T03-I M6-123	P-0021 448-T0 3-IM6		FGFR2 -PKP4	PKP 4	in-fr ame	fiv e_ pri me	123242 661	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - PKP4 (NM_003628) rearrangement: t(2;10)(q24.1 ;q26.13)(chr2:g.159395413:: chr10:g.123242661) Protein Fusion: in frame {FGFR2:PKP4}	NA
EVT-P-005 9669-T01-I M7-125	P-0059 669-T0 1-IM7		FGFR2 -POC1 B	PO C1B	in-fr ame	fiv e_ pri me	123242 798	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - POC1B (NM_001199777.2) fusion: t(10;12)(q26.13;q21.3 3)(chr10:g.123242798::chr12 :g.89857934) Protein Fusion: in frame {FGFR2:POC1B}	NA
EVT-P-000 8657-T01-I M5-126	P-0008 657-T0 1-IM5		FGFR2 -PPP1 R21	PPP 1R2 1	in-fr ame	fiv e_ pri me	123240 684	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR1 (NM_000141) - PPP1R21 (NM_152994) Rearrangement : t(2;10)(2p1 6.3;10q26.13)(chr2:g.487218 64::chr10:g.123240684) Protein fusion: in frame (FGFR2-KLRAQ1)	NA
EVT-P-005 7247-T02-I M7-127	P-0057 247-T0 2-IM7		FGFR2 -PRDX 3	PR DX3	in-fr ame	fiv e_ pri me	123242 894	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c.2301+318:FGFR2_c.36+28 7:PRDX3del Protein Fusion: in frame {FGFR2:PRDX3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 1394-T01-I M5-129	P-0011 394-T0 1-IM5		FGFR2 -RABG AP1L	RAB GA P1L	in-frame	five_prime	123240 065	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.123240065) Protein fusion: in frame (FGFR2-RABGAP1L)	NA
EVT-P-001 2639-T01-I M5-130	P-0012 639-T0 1-IM5		FGFR2 -SEPT1 N10	SEPTIN 10	unknown	five_prime	123240 798	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - SEPT10 (NM_144710) rearrangement: t(2;10)(q13;q26.13)(chr2:g.110325395::chr10:g.123240798) Protein fusion: mid-exon (FGFR2-SEPT10)	NA
EVT-P-005 5202-T01-I M6-131	P-0055 202-T0 1-IM6		FGFR2 -SH3GLB1	SH3GLB 1	in-frame	five_prime	123242 019	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - SH3GLB1 (NM_001206651) rearrangement: t(1;10)(p22.3;q26.13)(chr1:g.87191292::chr10:g.123242019) Protein Fusion: in frame {FGFR2:SH3GLB1}	NA
EVT-P-001 1516-T01-I M5-132	P-0011 516-T0 1-IM5		FGFR2 -SHC2	SHC2	in-frame	five_prime	123243 182	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - SHC2 (NM_012435) Rearrangement : t(10;19)(q26.13;p13.3)(chr10:g.123243182::chr19:g.447779) Protein fusion: in frame (FGFR2-SHC2)	NA
EVT-P-003 2152-T01-I M6-133	P-0032 152-T0 1-IM6		FGFR2 -SHTN1	SHTN1	in-frame	five_prime	123241 513	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2301+1699:FGFR2_c.534+751:KIAA1598 Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-005 7005-T01-I M6-134	P-0057 005-T0 1-IM6		FGFR2 -SHTN1	SHTN1	in-frame	five_prime	123239 901	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2302-366:FGFR2_c.534+660:KIAA1598del Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-006 4573-T01-I M7-135	P-0064 573-T0 1-IM7		FGFR2 -SKI	SKI	in-frame	five_prime	123242 232	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - SKI (NM_003036) fusion: t(1;10)(p36.33;q26.13)(chr1:g.2207811::chr10:g.123242232) Protein Fusion: in frame {FGFR2:SKI}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_st_atu_s	tar_get_r_ole	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 4244-T01-I M6-136	P-0054 244-T0 1-IM6		FGFR2 -SORBS1	SO RBS1	in-fr ame	fiv e_pri me	123243 201	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - SORBS1 (NM_001034954) fusion: c.2301+11:FGFR2_c. 1738-382:SORBS1del Protein Fusion: in frame {FGFR2:SORBS1}	NA
EVT-P-005 7277-T01-I M6-137	P-0057 277-T0 1-IM6		FGFR2 -SRRT	SR RT	out-of-fr ame	fiv e_pri me	123242 651	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - SRRT (NM_015908) fusion: t (7;10)(q22.1;q26.13)(chr7:g. 100475766::chr10:g.123242 651) Protein Fusion: out of frame {FGFR2:SRRT}	NA
EVT-P-005 5975-T01-I M6-139	P-0055 975-T0 1-IM6		FGFR2 -STAT4	STA T4	in-fr ame	fiv e_pri me	123240 312	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - STAT4 (NM_001243835) rearrangement: t(2;10)(q32.3 ;q26.13)(chr2:g.191936121:: chr10:g.123240312) Protein Fusion: in frame {FGFR2:STAT4}	NA
EVT-P-004 5185-T01-I M6-141	P-0045 185-T0 1-IM6		FGFR2 -TACC2	TAC C2	in-fr ame	fiv e_pri me	123240 014	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-479:FGFR2_c.8128- 2249:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-003 1242-T01-I M6-142	P-0031 242-T0 1-IM6		FGFR2 -TACC2	TAC C2	unk now n	fiv e_pri me	123239 432	18	802	False	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2405:FGFR2_c.7896+237: TACC2inv Protein Fusion: mid-exon {FGFR2:TACC2}	NA
EVT-P-002 3364-T03-I M6-143	P-0023 364-T0 3-IM6		FGFR2 -TACC2	TAC C2	unk now n	fiv e_pri me	123241 587	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) rearrangement: c.2301+1625 :FGFR2_c.34-5469:TACC2d up Antisense Fusion	NA
EVT-P-002 4187-T01-I M6-144	P-0024 187-T0 1-IM6		FGFR2 -TACC2	TAC C2	unk now n	fiv e_pri me	123240 573	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) Fusion (FGFR2 exon 17 with TACC2 exon12) : c.2302-10 38:FGFR2_c.8009:TACC2in v Protein Fusion: mid-exon {FGFR2:TACC2}	NA

event_id	sampl_e_id	pa_tie_nt_i_d	fusion_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_r_ole	breakp_o_int_ge_nomic	break_poi_nt_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 0962-T01-I M6-145	P-0050 962-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123241 844	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1368:FGFR2_c.7608 +332:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-003 6358-T01-I M6-146	P-0036 358-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123239 743	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-208:FGFR2_c.7344+ 468:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-005 7746-T01-I M6-149	P-0057 746-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123241 998	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128 -402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-005 7746-T02-I M6-150	P-0057 746-T0 2-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123241 998	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128 -402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-002 2337-T06-I M6-151	P-0022 337-T0 6-IM6		FGFR2 -TBC1 D4	TBC 1D4	in-fr ame	fiv e_ pri me	123241 723	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2) (chr10:g.123241723::chr13: g.75946350) Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-002 2337-T04-I M6-152	P-0022 337-T0 4-IM6		FGFR2 -TBC1 D4	TBC 1D4	in-fr ame	fiv e_ pri me	123241 723	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2) (chr10:g.123241723::chr13: g.75946350) Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-000 5291-T01-I M5-153	P-0005 291-T0 1-IM5		FGFR2 -TFEC	TFE C	unk now n	fiv e_ pri me	123241 729	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26. 13;q31.2)(chr10:g.12324172 9;chr7:g.115644112) Transcript fusion (FGFR2-TFEC)	NA

event_id	sampl_e_id	pa_tie_nt_i_d	fusion_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_r_ole	breakp_o_int_ge_nomic	break_poi_nt_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 6640-T01-I M6-154	P-0016 640-T0 1-IM6		FGFR2 -TTC2 8	TTC 28	in-fr ame	fiv e_ pri me	123241 944	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement : t(10;22)(q2 6.11;q12.1)(chr10:g.1232419 44::chr22:g.28666001) Protein Fusion: in frame {FGFR2:TTC28}	NA
EVT-P-005 1558-T01-I M6-155	P-0051 558-T0 1-IM6		FGFR2 -TTC2 8	TTC 28	in-fr ame	fiv e_ pri me	123243 158	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1) (chr10:g.123243158::chr22: g.28605071) Protein Fusion: in frame {FGFR2:TTC28}	NA
EVT-P-002 9726-T01-I M6-156	P-0029 726-T0 1-IM6		FGFR2 -UBP1	UBP 1	in-fr ame	fiv e_ pri me	123240 062	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - UBP1 (NM_001128161) fusion: t(3;10)(p22.3;q26.13)(chr3:g.33449450::chr10:g.12 3240062) Protein Fusion: in frame {FGFR2:UBP1}	NA
EVT-P-000 4089-T01-I M5-158	P-0004 089-T0 1-IM5		FGFR2 -VCL	VCL	unk now n	fiv e_ pri me	123242 749	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2(NM_000141) - VCL (NM_014000) Rearrangement : c.2301+46 3:FGFR2_c.1032:VCLinv Protein fusion: mid-exon (FGFR2-VCL)	NA
EVT-P-005 9238-T02-I M7-159	P-0059 238-T0 2-IM7		FGFR2 -WAC	WA C	in-fr ame	fiv e_ pri me	123242 066	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301 +1146:FGFR2_c.1437+679: WACinv Protein Fusion: in frame {FGFR2:WAC}	NA
EVT-P-005 8031-T01-I M6-160	P-0058 031-T0 1-IM6		FGFR2 -WAC	WA C	in-fr ame	fiv e_ pri me	123243 180	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301 +32:FGFR2_c.382-351:WAC inv Protein Fusion: in frame {FGFR2:WAC}	NA
EVT-P-002 5807-T01-I M6-161	P-0025 807-T0 1-IM6		FGFR2 -YTHD F3	YTH DF3	unk now n	thr ee_ pr im e	123242 317	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase		YTHDF3 (NM_152758) - FGFR2 (NM_000141) rearrangement: t(8;10)(q12.3 ;q26.13)(chr8:g.64081967::c hr10:g.123242317) Protein Fusion: mid-exon {YTHDF3:FGFR2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 6399-T01-I M6-162	P-0056 399-T0 1-IM6		FGFR2 -ZFAT	ZFAT	in-frame	five_prime	123240 282	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - ZFAT (NM_020863) rearrangement: t(8;10)(q24.2;q26.13)(chr8:g.135564146::chr10:g.123240282) Protein Fusion: in frame {FGFR2:ZFAT}	NA
EVT-P-004 2241-T01-I M6-163	P-0042 241-T0 1-IM6		FGFR2 -ZMYM4	ZMYM4	in-frame	five_prime	123242 889	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - ZMYM4 (NM_005095) fusion: t(1;10)(p34.3;q26.13)(chr1:g.35833702::chr10:g.123242889) Protein Fusion: in frame {FGFR2:ZMYM4}	NA
EVT-P-001 6195-T01-I M6-164	P-0016 195-T0 1-IM6		GAB2- FGFR2	GAB2	in-frame	five_prime	123242 812	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(10;11)(q26.11;q14.1)(chr10:g.123242812::chr11:g.78030553) Protein Fusion: in frame {FGFR2:GAB2}	NA
EVT-P-001 3756-T01-I M5-165	P-0013 756-T0 1-IM5		INA-F GFR2	INA	in-frame	three_prime	123242 695	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		INA (NM_032727) - FGFR2 (NM_000141) Rearrangement : c.1066-3500:INA_c.2301+517:FGFR2inv Protein fusion: in frame {INA-FGFR2}	NA
EVT-P-001 3756-T03-I M6-166	P-0013 756-T0 3-IM6		INA-F GFR2	INA	in-frame	three_prime	123242 695	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500:INA_c.2301+517:FGFR2inv Protein Fusion: in frame {INA:FGFR2}	NA
EVT-P-000 2810-T01-I M3-169	P-0002 810-T0 1-IM3		KIAA1 217-F GFR2	KIAA1217	in-frame	five_prime	123240 874	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR2-KIAA1217)	NA

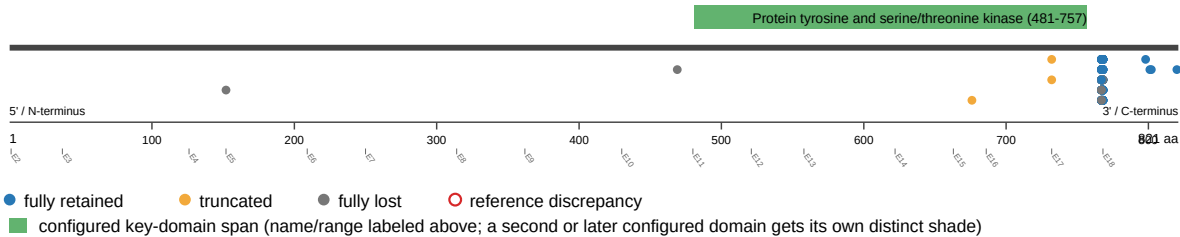
event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 0386-T02-I M5-170	P-0000 386-T0 2-IM5		MARV ELD3- FGFR2	MA RVE LD3	unk now n	fiv e_ pri me	123243 101	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13) (chr16:g.71660650::chr10:123243101) Antisense fusion	NA
EVT-P-002 4094-T01-I M6-172	P-0024 094-T0 1-IM6		MYH9- FGFR2	MY H9	in-fr ame	fiv e_ pri me	123240 418	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240418::chr22:g.36682296) Protein Fusion: in frame {FGFR2:MYH9}	NA
EVT-P-003 2208-T01-I M6-173	P-0032 208-T0 1-IM6		MYO1 8B-FG FR2	MY O18 B	in-fr ame	fiv e_ pri me	123241 929	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929::chr22:g.26280218) Protein Fusion: in frame {FGFR2:MYO18B}	NA
EVT-P-002 1975-T01-I M6-175	P-0021 975-T0 1-IM6		NID2-F GFR2	NID 2	out- of-fr ame	thre e_ pr im e	123240 637	18	768	True	lost	0.0	False		Prote in tyr osine and se rine/ thr eonine ki nase		NID2 (NM_007361) - FGFR2 (NM_000141) rearrangement: t(10;14)(q26.13;q22.1)(chr10:g.123240637::chr14:g.52482743) Protein Fusion: out of frame {NID2:FGFR2}	NA
EVT-P-001 2423-T01-I M5-176	P-0012 423-T0 1-IM5		NOL4- FGFR2	NOL 4	in-fr ame	fiv e_ pri me	123242 054	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - NOL4 (NM_003787) Rearrangement: t(10;18)(q26.13;q12.1)(chr10:g.123242054::chr18:g.31563418) Protein fusion: in frame (FGFR2-NOL4)	NA
EVT-P-002 3103-T01-I M6-177	P-0023 103-T0 1-IM6		NOL4- FGFR2	NOL 4	in-fr ame	fiv e_ pri me	123241 921	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused to NOL4 exons 10-11): t(10;18)(q26.13;q12.1)(chr10:g.123241921::chr18:g.31518701) Protein Fusion: in frame {FGFR2:NOL4}	NA
EVT-P-000 8693-T01-I M5-178	P-0008 693-T0 1-IM5		NRAP- FGFR2	NR AP	in-fr ame	fiv e_ pri me	123242 277	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase			FGFR2 (NM_000141) - NRAP (NM_198060) rearrangement: c.2301+935: FGFR2_c.2458-427:NRAPdel Protein fusion: in frame (FGFR2-NRAP)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 6328-T01-I M5-179	P-0006 328-T0 1-IM5		NRBF2-FGFR2	NRBF2	in-frame	five_prime	123241245	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - NRBF2 (NM_030759) rearrangement: c.2302-1710: FGFR2_c.157-364:NRBF2inv Protein fusion: in frame (FGFR2-NRBF2)	NA
EVT-P-004 9071-T01-I M6-180	P-0049 071-T0 1-IM6		NRG3-FGFR2	NRG3	out-of-frame	three_prime	123244882	16	732	True	disrupted	0.0938628 15884476 54	True			Protein tyrosine and serine/threonine kinase	NRG3 (NM_001010848) - FGFR2 (NM_000141) rearrangement: c.823+108972:NRG3_c.2195+27:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-002 2337-T01-I M6-181	P-0022 337-T0 1-IM6		NRG3-FGFR2	NRG3	out-of-frame	three_prime	123241723	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		NRG3 (NM_001010848) - FGFR2 (NM_000141) Rearrangement : c.953+89877:NRG3_c.2301+1489:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-003 2171-T03-I M6-182	P-0032 171-T0 3-IM6		OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254: FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN}	NA
EVT-P-003 2171-T01-I M6-183	P-0032 171-T0 1-IM6		OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254: FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN}	NA
EVT-P-002 2781-T01-I M6-184	P-0022 781-T0 1-IM6		PDZRN4-FGFR2	PDZRN4	in-frame	five_prime	123241997	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) Protein Fusion: in frame {FGFR2:PDZRN4}	NA
EVT-P-004 4768-T01-I M6-187	P-0044 768-T0 1-IM6		RNF145-FGFR2	RNF145	out-of-frame	three_prime	123243127	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		RNF145 (NM_144726) - FGFR2 (NM_000141) rearrangement: t(5;10)(q33.3;q26.13)(chr5:g.158598146::chr10:g.123243127) Protein Fusion: out of frame {RNF145:FGFR2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 6235-T01-I M5-188	P-0006 235-T0 1-IM5		ROCK 1-FGFR R2	ROCK1	in-frame	five_prime	123240 133	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - ROCK1 (NM_005406) rearrangement: t(10;18)(q26.13;q11.1)(chr10:g.123240133::chr18:g18680051) Protein fusion: in frame (FGFR2-ROCK1)	NA
EVT-P-003 9370-T01-I M6-189	P-0039 370-T0 1-IM6		RPAP3 -FGFR 2	RPAP3	in-frame	three_prime	123242 832	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		RPAP3 (NM_024604) - FGFR2 (NM_000141) rearrangement: t(10;12)(q26.13;q13.11)(chr10:g.123242832::chr12:g.48092574) Protein Fusion: in frame {RPAP3:FGFR2}	NA
EVT-P-002 5074-T01-I M6-190	P-0025 074-T0 1-IM6		SHTN1 -FGFR 2	SHTN1	in-frame	five_prime	123240 122	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) Fusion (FGFR2 exon 17 with KIAA1598 exon 5): c.2302-587:FGFR2_c.267+2596:KIAA1598 Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-003 2507-T01-I M6-191	P-0032 507-T0 1-IM6		SORBS1-FGFR2	SORBS1	unknown	five_prime	123242 268	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) Rearrangement : c.2301+944:FGFR2_chr10:g.97250518del Transcript Fusion {FGFR2:SORBS1}	NA
EVT-P-002 2337-T01-I M6-192	P-0022 337-T0 1-IM6		TBC1D4-FGFR2	TBC1D4	in-frame	five_prime	123241 723	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - TBC1D4 (NM_014832) Rearrangement : t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946348) Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-002 1788-T01-I M6-193	P-0021 788-T0 1-IM6		VPS26A-FGFR2	VPS26A	in-frame	three_prime	123242 226	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		VPS26A (NM_004896) - FGFR2 (NM_000141) Rearrangement : c.4-1035:VPS26A_c.2301+986:FGFR2inv Protein Fusion: in frame {VPS26A:FGFR2}	NA

domain retention outliers

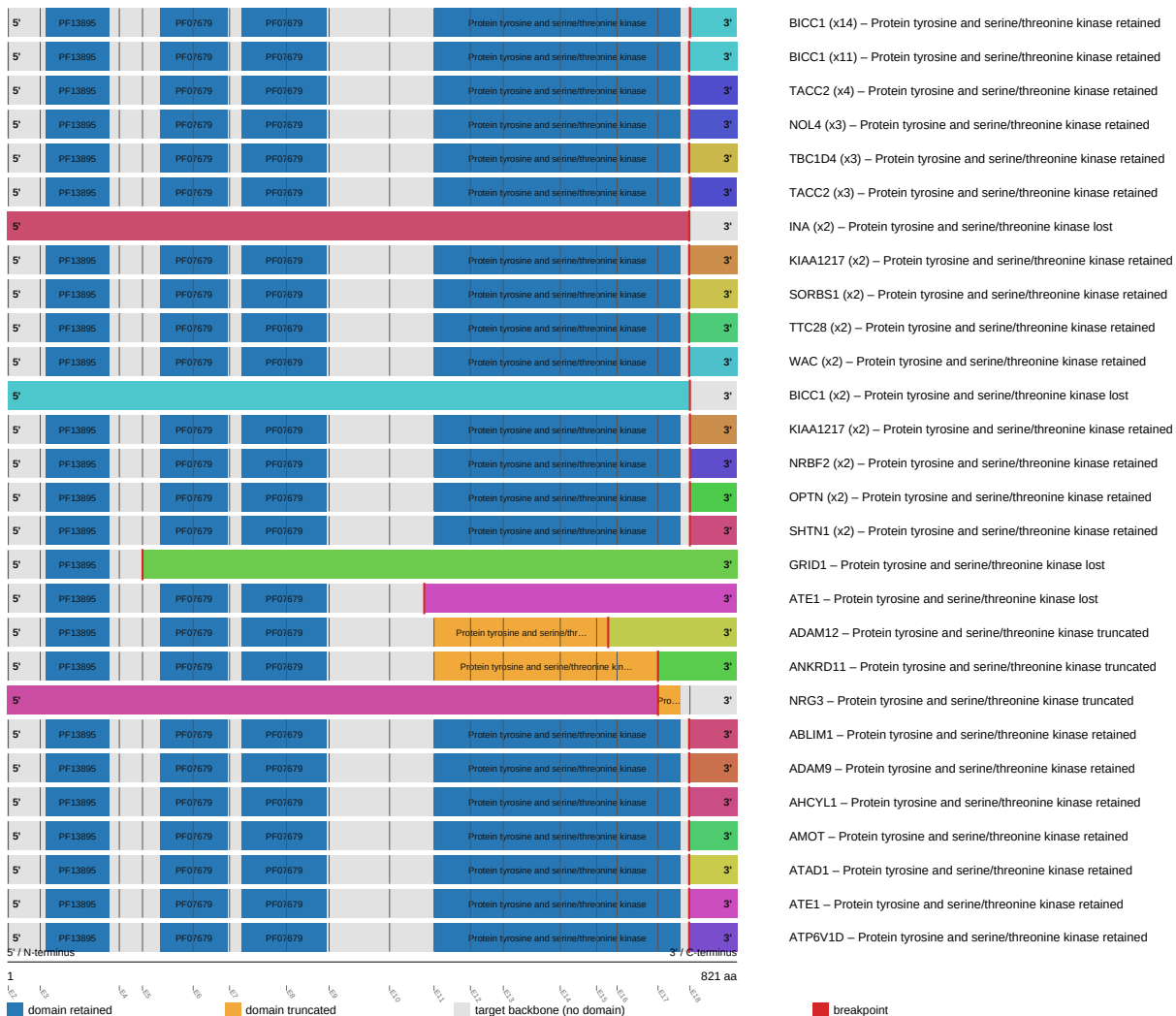
FGFR2 domain-retention track



fusion schematic

FGFR2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR2 portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 92 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

